Magnetic Skyrmions

A LaTeX presentation template for Utrecht University

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June 17, 2026

Outline

Background

$$H = H_J + H_D$$

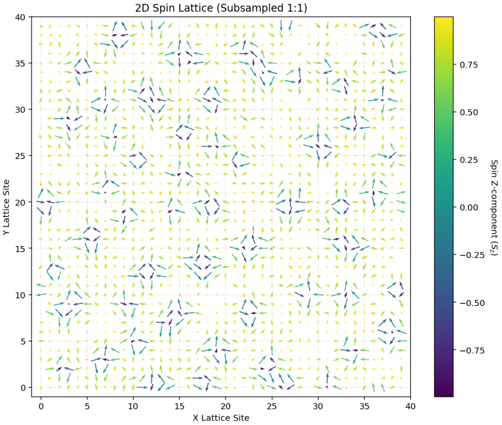
Heisenberg exchange Hamiltonian : $H_J = -J\sum_r \vec{S}_r \cdot \left(\vec{S}_{r+\hat{x}} + \vec{S}_{r+\hat{y}}\right) - H_Z \sum_r \vec{S}_r$

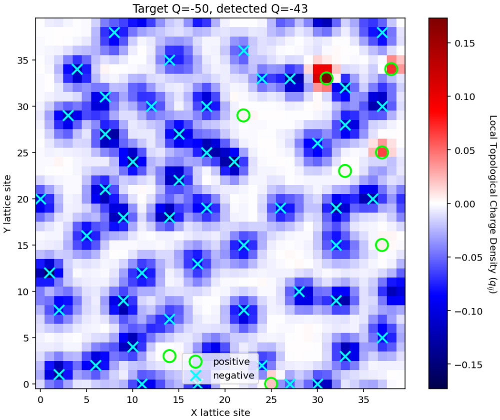
Dzyaloshinskii-Moriya interaction Hamiltonian : $H_D = D\sum_r \vec{S}_r \times \vec{S}_{r+\hat{x}} \cdot \hat{x} + \vec{S}_r \times \vec{S}_{r+\hat{y}} \cdot \hat{y}$

Simulation parameters

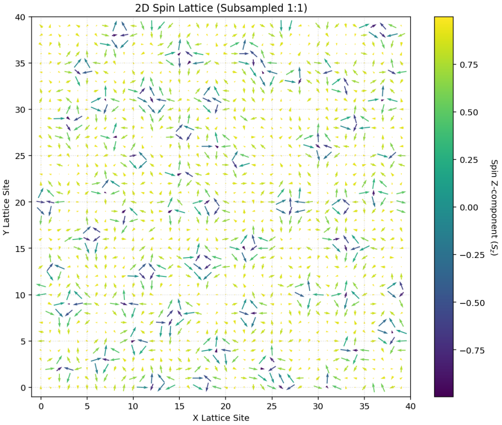
40 x 40 grid
H, D being variable and J constant
beta being quenched

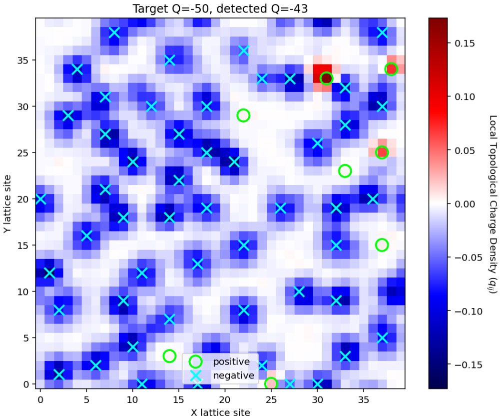
Spin lattice



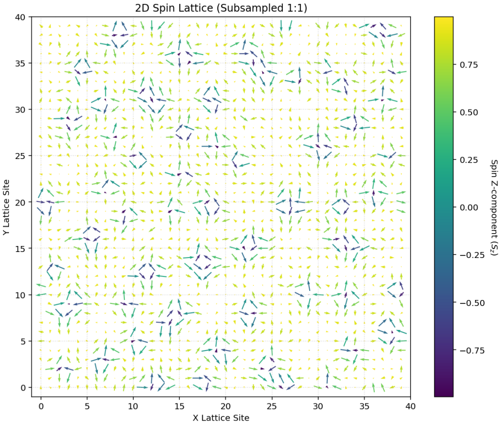


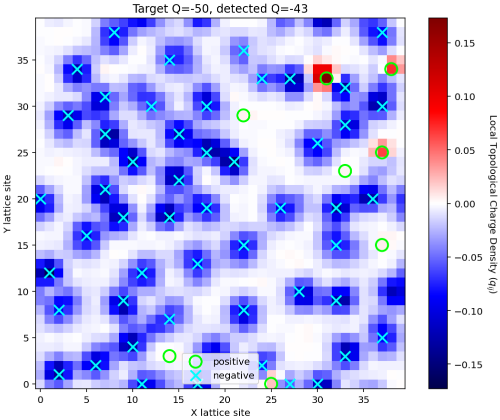
Wrapping number





Skyrmion detection





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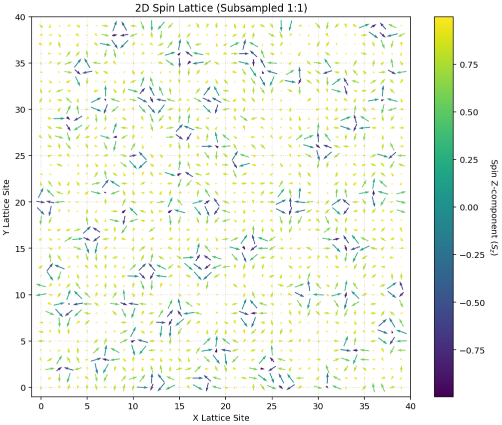
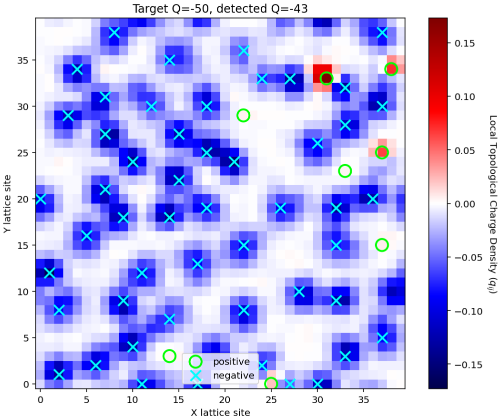
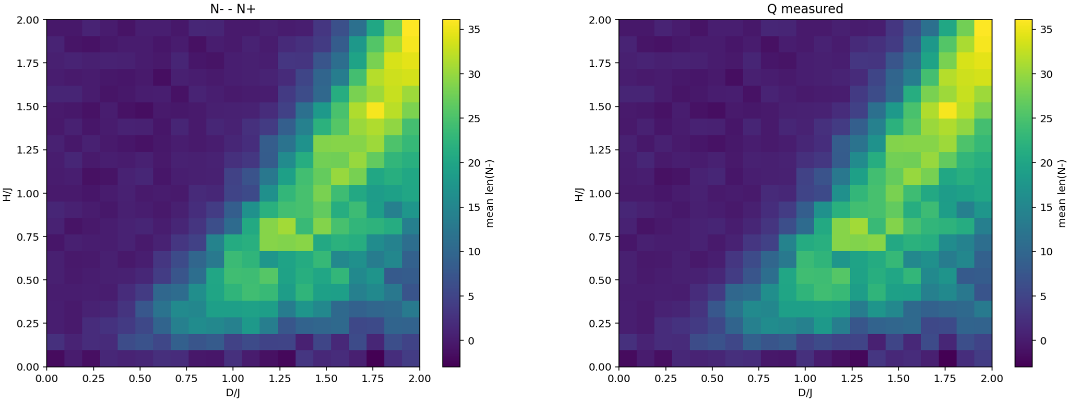
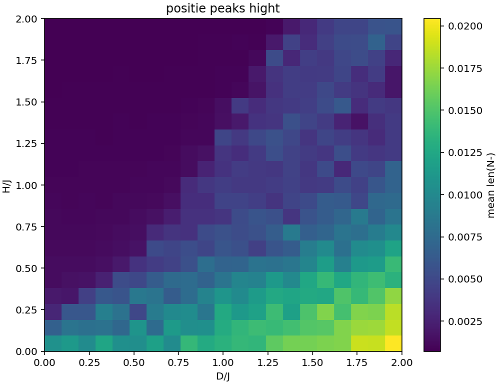


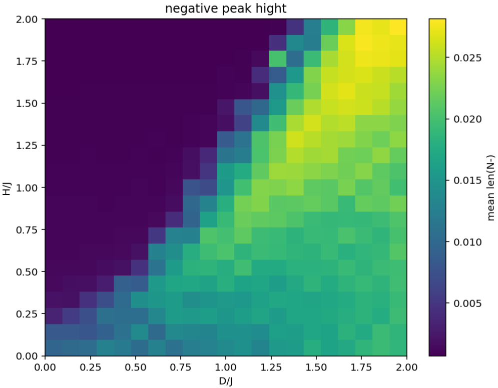


Phase Diagram

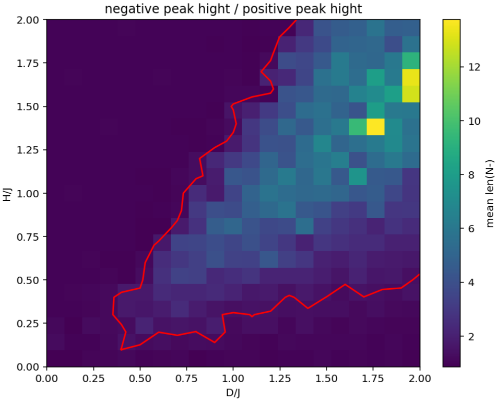


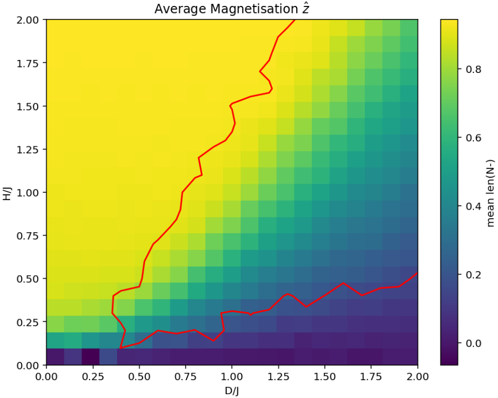
Phase Separation 1





Phase Separation 2





Temperature Dependence